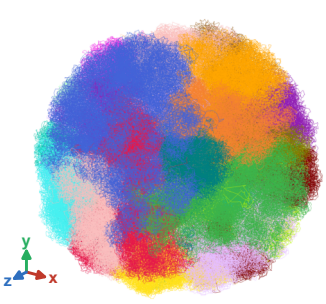
single-cell reconstruction at high resolution (EMber, homolog-resolved)

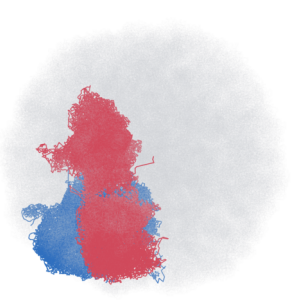
a **EMber deep (U-Chrom): a diploid human genome at 5 kb**

GM12878 scMicro-C (Wu 2025), deepest cell: 3.01 M contacts, 1.16 M particles (2 × 23 chains); 10 models in 11.8 min on one RTX 2080 Ti

run-to-run r = 0.98 per copy chain at 5 kb (authors' hickit runs 0.98); 37 % of contacts phased and used; test, 7 held-back cells: 20 kb r median 0.98 (min 0.97), held-out AUC vs NucDynamics +0.012

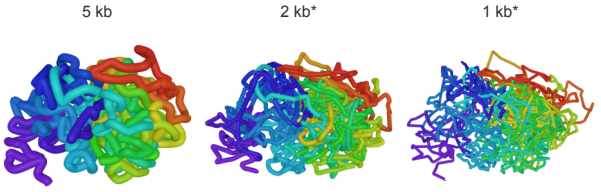


coloured by chromosome



chr1 copies: paternal maternal

chr1:53–56 Mb, paternal copy; rainbow along the region

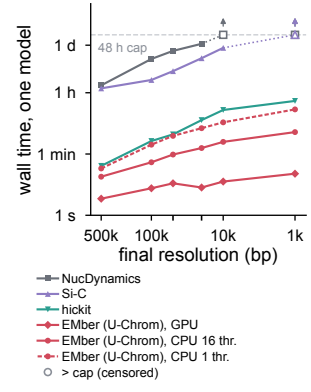


one model each, superposed on the 5 kb model
* exploratory

single-cell reconstruction: EMber vs existing methods, one benchmark

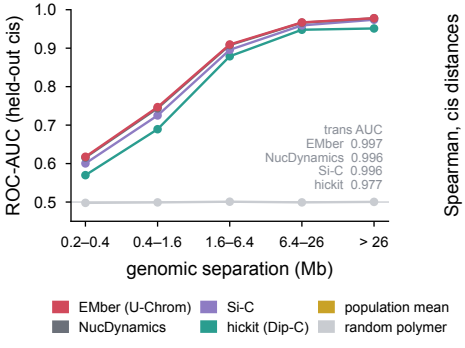
b **Time for one model**

Cell 1; CPU: EPYC 9754, GPU: H100



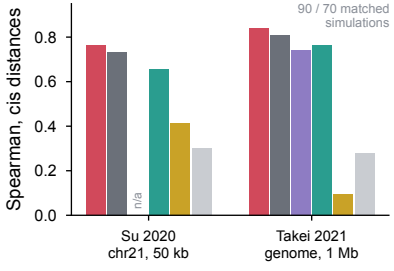
c **Held-out contacts**

8 cells × 3 folds; matched-separation negatives



d **Imaging truth (simulated Hi-C)**

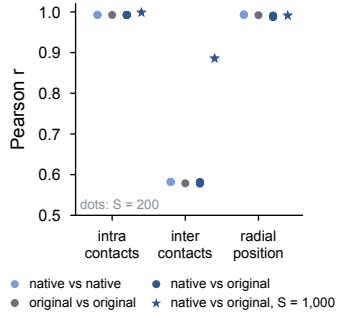
median model vs imaged distances



population reconstruction (IGM)

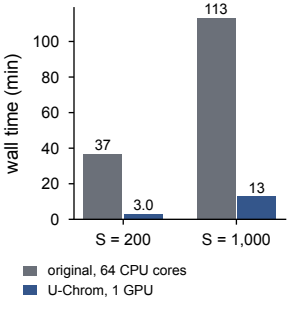
e **IGM: native vs original**

WTC11 2 Mb; population agreement



f **IGM speed**

same input and schedule



g **How many structures?**

WTC11 2 Mb, 3 seeds; grey: minutes on 1 GPU

